NEURALPATH CHEATSHEET

# Module 3: Deep Learning Core

Understand perceptrons, MLPs, activations, loss landscapes, reverse-mode backpropagation, AdamW

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## Core Formulas, Mental Models & Engineering Rules

- Backpropagation computes exact gradients for every weight in one reverse pass.
- Modern Activations: ReLU for CNNs, GELU or SwiGLU for Transformers.
- Init: Kaiming/He for ReLU; Xavier/Glorot for symmetric Tanh/Sigmoid.
- Normalization: LayerNorm or RMSNorm stabilizes deep residual stacks independent of batch size.
- Optimizer default: AdamW (decoupled weight decay) with cosine learning-rate warmup + decay.

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Estimated study time: 5 hours · Badge: Neural Architect
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